Shiv Nadar University, Chennai

Department of Information Technology

25IT1731 Computational Thinking and Problem Solving with Python

Ex 5 – List and Tuple

Part A

1	Write a Python program to generate a list containing numbers between 1 and 20 that are divisible by 2 or 3. Perform the following operations on the generated list: a) Display the complete list. b) Display the first and last elements. c) Display the first three elements. d) Ask the user to enter a number and check whether it is present in the list. e) Ask the user to enter a number and display its position if it is present. f) Add the number 18 to the list. g) Insert the number 12 at the second position. h) Find how many times 12 occurs in the list. i) Remove one occurrence of 12 from the list. j) Remove the last element and display the removed element. k) Reverse the list. l) Arrange the elements in ascending order. m) Display the elements in descending order. n) Find and display the smallest value. o) Find and display the largest value. p) Calculate and display the total of all the elements. q) Calculate and display the average of the elements. r) Display the total number of elements in the list.
2	A supermarket scanner records the product ID every time an item is scanned. Sometimes the same product is scanned more than once. Consider the following list: products = [101, 102, 103, 101, 104, 102, 105, 101] Write a Python program to create a new list containing only the product IDs that occur more than once. Each repeated product ID should appear only once in the new list.
3	Write a Python program to read 10 integers from the user and store them in a list. Write separate user-defined functions to display the list, find the smallest element, find the largest element, and search for an element. Ask the user to enter an element to search for and display its position if it is present in the list; otherwise, display an appropriate message. Do not use built-in functions

4	a) A data analyst wants to generate different groups of numbers from a given range. Write a Python program using list comprehension to create the following lists: • A list containing numbers from 1 to 50. • A list containing only the even numbers from 1 to 50. b) An online store maintains the prices of products in a list. Write a Python program using list comprehension to perform the following tasks. prices = [250, 799, 1200, 450, 999, 1500, 350, 2000] • Create a list containing only products priced above ₹1000. • Create a list containing prices after applying a 10% discount.
5	A class has five students. The marks obtained by each student in three subjects are stored as a nested list, where each inner list represents the marks of one student. marks = [[78, 85, 91], [62, 74, 69], [88, 92, 95], [45, 67, 58], [90, 81, 87]] Write a Python program to display the marks of each student, calculate and display the total marks obtained by each student, and calculate and display the average marks obtained by each student.
6	Write a Python program to read two matrices from the user and perform matrix addition and matrix multiplication. Write separate user-defined functions to read a matrix, display a matrix, add two matrices, and multiply two matrices. Display the two input matrices and the corresponding resultant matrices for both addition and multiplication. The program should check whether the given matrices are suitable for the required operation before performing it.
7	Given the following 3 × 2 matrix, write a Python program to find and display its transpose using list comprehension. The program should display both the original matrix and the resultant transpose. matrix = [[1, 2], [3, 4], [5, 6]] Expected Output: [1, 3, 5] [2, 4, 6]
8	An online shopping application stores the details of a product in a tuple. The product information includes the product name, category, price, and rating. Write a Python program to create the following tuple: product = ("Wireless Mouse", "Electronics", 799, 4.5) Perform the following operations: a) Display the product name and price using tuple indexing. b) Display the category and rating using tuple slicing. c) Ask the user to enter a value and check whether it is present in the tuple.

	d) Display the number of elements in the tuple. e) Unpack the tuple into separate variables and display the product details as a meaningful sentence. f) Create another tuple containing the discount percentage and stock availability, and concatenate it with the product tuple to create a new tuple.
9	Write a function that accepts a tuple of numbers and creates two separate tuples: one containing the elements at even positions and another containing the elements at odd positions. Return both tuples as a nested tuple. Sample Input: (10, 20, 30, 40, 50, 60, 70) Expected Output: ((10, 30, 50, 70), (20, 40, 60))
10	A railway application stores the schedule of trains operating from a station using a nested tuple. Each inner tuple contains the train number, train name, destination, and departure time. trains = ((12601, "Chennai Express", "Bengaluru", "06:00"), (12602, "Coastal Express", "Hyderabad", "08:30"), (12603, "City Express", "Coimbatore", "10:15"), (12604, "Intercity Express", "Madurai", "14:00")) Write a Python program to perform the following operations: a) Display the train name and destination of the first train. b) Display the complete details of the third train using indexing. c) Unpack the details of each train into separate variables and display them in a meaningful format. d) Ask the user to enter a destination and display the details of the train travelling to that destination. e) Display the train numbers of all trains using the nested tuple. f) Display the total number of trains in the schedule.

Part B

11	A teacher has a list of marks and wants to display them in ascending order. Write a Python program to sort the marks without using sort() or sorted(). Hint: use bubble sort
12	Two classes have registered students for two different workshops: workshop1 = ["Arun", "Priya", "Rahul", "Meena"] workshop2 = ["Priya", "Meena", "Kiran", "Divya"] Write a Python program to find the students who registered for Workshop 1 but not Workshop 2.
13	A shop records the sales made by four branches over three months using a nested list. The sales information is given below: sales = [[25000, 30000, 28000, 32000], [27000, 29000, 31000, 35000],

	[30000, 33000, 29000, 36000]] Write a Python program to display the sales recorded for each month, determine the sales made by the second branch in the first month, calculate the total sales for each month, and find the highest individual sales value.
14	Write a Python function that takes a tuple and an integer k, and rotates the elements of the tuple to the right by k positions. Sample Input: my_tuple = (1, 2, 3, 4, 5), k = 2 , Expected Output: (4, 5, 1, 2, 3)
15	Given a sentence, split it into words, and write a list comprehension that keeps only the words with a length greater than 3 characters, converting those surviving words to uppercase. Sample Input: text = "Python is an incredible and versatile programming language", Expected Output: ['PYTHON', 'INCREDIBLE', 'VERSATILE', 'PROGRAMMING', 'LANGUAGE']
16	Generate a list of coordinate tuples (x, y) where x ranges from 1 to 3 and y ranges from 1 to 3, but exclude any coordinates where the sum of x and y is an odd number. Sample Input: xvals = [1, 2, 3], yvals = [1, 2, 3] , Expected Output: [(1, 1), (1, 3), (2, 2), (3, 1), (3, 3)]
17	Write a function that accepts a tuple of elements and returns a new tuple with all adjacent duplicate values collapsed into a single instance (preserving the relative order). Sample Input: (1, 1, 2, 3, 3, 3, 4, 2, 2), Expected Output: (1, 2, 3, 4, 2)

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